Youngkwon Ko

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01 Education

Yonsei University, Seoul, Republic of Korea March 2023 - Present
B.S. in Artificial Intelligence
Currently on a leave of absence

02 Publications

01 **A Mission-Aware Autoencoder Bottleneck Compression Framework for Drone-Based Hyperspectral Image Transmission.**

Youngkwon Ko, Yunbo Shim, Doun Lee, and Seayul Choi.

Marine Science & Technology, 2026.

02 **Provenance-Preserving Schema and Resource Construction for Information Extraction from Historical Battle Records: A Corpus Study of the Inchon Landing.**

Yunbo Shim, **Youngkwon Ko**, et al.

Proceedings of the Korea Computer Congress (KCC 2026), 2026.

03 Research & professional experience

Republic of Korea Navy December 2025 - August 2027 (expected)
Artificial Intelligence Specialist

- Conducted applied AI research and development as an AI-specialized enlisted service member.
- Developed a mission-aware hyperspectral image transmission framework combining autoencoder-based compression with spectral patch prioritization.
- Achieved approximately 613× and 710.5× latent-payload reductions on KSC and Salinas, respectively, while outperforming random and edge-based selection on important-region retrieval metrics.
- Participating in technical research and discussions on hyperspectral satellite systems in collaboration with the Agency for Defense Development (ADD).

Creai+it September 2024 - December 2025
Co-Leader, AI Startup Club

- Led AI product development projects spanning document intelligence, legal AI, assistive technology, and generative AI.
- Developed projects including *Somni*, an AI-assisted narrative development system, and *Miney*, a legal document analysis and case-retrieval system.
- Collaborated with Upstage on a Solar-based intelligent document processing system and participated in the 2025 AGI Agent Hackathon.

Research & professional experience

Root Education September 2025 - December 2025

Co-Founder

- Co-founded an AI-powered education platform that organized admissions-focused influencer content into a structured learning environment.
- Developed AI-assisted learning features and contributed to product design and frontend development using Next.js and TypeScript.

Little Samo September 2025 - October 2025

Contract Software Developer

- Developed *SamoAI-InstagramTracker*, an AI-powered Instagram influencer discovery and data collection application built on the SamoAI multi-agent framework.
- Contributed to maintenance of the SamoAI engine and CLI-based agent chat interface.

Yonsei Medisys Lab January 2025 - June 2025

Research Intern

- Conducted low-dose CT image quality assessment research using 11,200 CT images with MOS quality annotations.
- Trained and evaluated EfficientNet, ViT, TinyViT, and ManiQA models for medical image quality assessment.
- Investigated CT denoising fine-tuning strategies using an image quality assessment model as a reward signal.

04 Awards & honors

1st Place, Navy & Marine Corps Artificial Intelligence Competition 2026

RED S Design Award - 1st Prize June 2025

Interactive Media Design, Yonsei University

05 Technical skills

Programming	Python, TypeScript, C++, JavaScript
Machine learning	PyTorch, TensorFlow, OpenCV
Development	React, Next.js, Git, Linux